

Suricata Rules Explore

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Suricata Installation, Configuration & Rules Research

Installing Suricata (Ubuntu-Suricata)

bash

```
sudo apt install software-properties-common -y  
sudo add-apt-repository ppa:oisf/suricata-stable -y  
sudo apt update  
sudo apt install suricata -y
```

Why: Ubuntu's default repositories often carry an outdated Suricata version; adding OISF's official PPA ensures installing a current, actively maintained release.

bash

```
suricata --build-info
```

Why: confirms the install succeeded and shows version/build details.

Configuring Suricata

bash

```
sudo nano /etc/suricata/suricata.yaml
```

Edited two critical settings:

- `HOME_NET: "[192.168.93.0/24]"` — tells Suricata which address range counts as "internal network," which many rules use to distinguish internal vs. external traffic direction

- `interface: ens34` (under `af-packet:`) — tells Suricata which network card to actually capture traffic from; **must** be the VMnet1 interface (the shared lab network), not the NAT one, otherwise Suricata would only see the VM's own internet browsing traffic instead of the actual attacker ↔ target traffic

Downloading Detection Rules

bash

```
sudo suricata-update
```

Why: the Suricata engine itself contains no built-in "known bad" signatures — rules are a separate, downloadable component. This command fetches the ruleset (defaulted automatically to **Emerging Threats Open**, confirmed by its own log output: *"No sources configured, will use Emerging Threats Open"*) and compiles it into `/var/lib/suricata/rules/suricata.rules`. First run loaded ~67,948 rules, ~52,012 enabled.

Validating Config Before Running Live

bash

```
sudo suricata -T -c /etc/suricata/suricata.yaml -i ens34
```

Why: `-T` performs a "test only" run — checks the config file for syntax errors without starting actual packet capture, catching mistakes cheaply before a live run.

Running Suricata

bash

```
sudo suricata -c /etc/suricata/suricata.yaml -i ens34 -D
```

Why: starts Suricata for real, watching `ens34` live, `-D` runs it as a background daemon so it keeps running independent of the terminal session.

bash

```
ps aux|grep suricata
```

Why: confirms the process is actually running (visible with its PID), separate from checking via `systemctl` (which initially showed a stale failed state from an earlier config attempt before the interface/HOME_NET fix).

Watching Alerts Live

bash

```
sudo tail -f /var/log/suricata/fast.log
```

Why: `fast.log` is a plain-text, alerts-only log (unlike `eve.json`, which logs everything including non-alert traffic metadata) — good for a quick, human-readable running view of only the actual detections as they happen.

First Real Detections

Ran `nmap -sV 192.168.93.139` from Kali → immediately saw `ET SCAN Possible Nmap User-Agent Observed` (sid:2024364) appear in `fast.log`, confirming the full pipeline (capture → rule match → alert → log) worked end-to-end.

Rules & Config Deep-Dive Research

bash

```
sudo ls -la /var/lib/suricata/rules/
```

Why: confirms exactly which files exist in the active rules directory — found `suricata.rules` (the compiled ruleset) and `classification.config`.

bash

```
sudo wc -l /var/lib/suricata/rules/suricata.rules
```

Why: counts total lines/rules currently loaded (~67,948), giving a concrete scale reference.

bash

```
sudo grep "2024364" /var/lib/suricata/rules/suricata.rules
```

Why: pulls the exact, full text of the specific rule that had been firing on our Nmap tests — used this as the concrete example to dissect every keyword (action, protocol, direction, `content`, `http_user_agent`, `classtype`, `sid`, `rev`, `metadata`) against the official Suricata rules documentation.

bash

```
sudo grep "^alert" /var/lib/suricata/rules/suricata.rules | shuf -n 5
```

Why: pulls 5 random rules to see variety beyond the one known example — surfaced rules spanning malware C2 detection, SQL injection, phishing, and adware, dated from 2006 to 2026, demonstrating the ruleset's real breadth and active maintenance.

bash

```
sudo grep -A3 "default-rule-path" /etc/suricata/suricata.yaml | sudo grep -A5 "rule-files" /etc/suricata/suricata.yaml
```

Why: confirms, directly from the config file, exactly where Suricata is told to load rules from (`/var/lib/suricata/rules/suricata.rules`) — connecting "what the config says" to "what we already found on disk."

bash

```
suricata-update list-sources
```

Why: reveals every rule source available in the wider ecosystem, not just the one currently active — showed free options (Abuse.ch feeds, community GitHub rules, OISF's own traffic-ID ruleset) alongside paid/commercial ones (Stamus Networks,

Secureworks, ET Pro), demonstrating that Suricata's detection capability is built on a modular "marketplace" of specialized threat-intelligence providers.

bash

```
sudo grep -A5 "eve-log" /etc/suricata/suricata.yaml | head -20
```

Why: confirms the config block controlling `eve.json` — `enabled: yes`, `filetype: regular`, `filename: eve.json` — explaining exactly why/how that structured log file has been continuously populated throughout testing.

Key concepts documented from this research

- **SID (Signature ID)** is a globally unique identifier assigned by whoever authored the rule (Emerging Threats, in our case) — separate from a CVE number, though often linked via the `reference:` field
- **SID numbering convention:** 1–1,000,000 historically Snort's own; 2,000,000+ used by Emerging Threats; custom/local rules conventionally start at 1,000,000+ to avoid collisions
- SIDs shown locally match exactly what's published on Emerging Threats' own public lookup portal (`threatintel.proofpoint.com/sid/<number>`), confirmed by direct side-by-side comparison
- Rules can be tagged with **MITRE ATT&CK** framework identifiers (`mitre_tactic_id`, `mitre_technique_id`), linking Suricata's detections to an industry-standard attacker-behavior taxonomy

Full detection pipeline documented (traced end-to-end using the Nmap example)

```
Kali sends traffic → arrives on ens34 (promiscuous mode) → Suricata decodes packet (IP/TC P/HTTP layers) → extracts HTTP User-Agent field → fast_pattern content match ("Nmap") used as quick filter → full rule conditions evaluated (flow state, exact content match) → match confirmed → alert action triggered → written simultaneously to fast.log (summary) and eve.json (full detail) → dashboard/tail -f displays result
```

```
Home kali linux-2025.4-vmware... Ubuntu-Berlin Ubuntu-Suricata  
us@ubuntu:~  
Session Actions Edit View Help  
To see these additional updates run: apt list --upgradable  
5 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Thu Jul 16 06:27:08 2026 from 192.168.93.130  
us@ubuntu:~$ sudo dpkg-reconfigure wireshark-common  
[sudo: authenticate] Password:  
sudo: Authentication failed, try again.  
[sudo: authenticate] Password:  
us@ubuntu:~$ sudo usermod -sG wireshark us  
us@ubuntu:~$ sudo suricata -c /etc/suricata/suricata.yaml -i ens34 -D  
I: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode  
E: pidfile: pid file '/var/run/suricata-pid' exists and Suricata appears to be running. Aborting!  
us@ubuntu:~$ ps aux | grep suricata  
suricata 1995 27.4 13.0 1107172 449080 ? Ssl 06:38 0:18 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata-pid --user suricata --group suricata  
us 2500 0.0 0.0 6716 2664 pts/0 S+ 06:39 0:00 grep --color=auto suricata  
us@ubuntu:~$ sudo tail -f /var/log/suricata/fast.log  
07/15/2026-12:16:27.405473 [**] [1:2035109:2] ET EXPLOIT Possible Citrix Application Delivery Controller Arbitrary Code Execution Attempt (CVE-2019-19781) M4 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48438 → 192.168.93.139:80  
07/15/2026-12:16:27.405473 [**] [1:2035110:2] ET EXPLOIT Citrix Application Delivery Controller Arbitrary Code Execution Attempt Scanner Attempt (CVE-2019-19781) [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48438 → 192.168.93.139:80  
07/15/2026-12:16:27.436237 [**] [1:2030469:7] ET EXPLOIT F5 TMUI RCE vulnerability CVE-2020-5902 Attempt M1 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48438 → 192.168.93.139:80  
07/15/2026-12:16:27.509557 [**] [1:2030483:4] ET EXPLOIT F5 TMUI RCE vulnerability CVE-2020-5902 Attempt M2 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48438 → 192.168.93.139:80  
07/15/2026-12:16:27.528324 [**] [1:2221015:1] SURICATA HTTP Host header ambiguous [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.93.130:48438 → 192.168.93.139:80  
07/15/2026-12:16:27.593690 [**] [1:2030585:1] ET EXPLOIT Cisco ASA/Firepower Unauthenticated File Read (CVE-2020-3452) M3 [**] [Classification: Attempted User Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48454 → 192.168.93.139:80  
07/15/2026-12:16:27.593690 [**] [1:2034262:1] ET EXPLOIT Cisco ASA and Firepower Path Traversal Vulnerability M1 (CVE-2020-3452) [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:48454 → 192.168.93.139:80  
07/15/2026-12:27:11.314705 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/15/2026-12:42:11.351953 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-06:32:20.015796 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-06:40:34.339739 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.93.130:46792 → 192.168.93.139:80  
07/16/2026-06:40:34.337442 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1]
```

```
Ubuntu 26.04 LTS ubuntu:~$ ssh -t ubuntu@192.168.23.160
Try contacting this VM's SSH server via 'ssh -t ubuntu@192.168.23.160' from host.

ubuntu@ubuntu:~$ ssh -t ubuntu@192.168.23.160
ubuntu:~$ ssh -t ubuntu@192.168.23.160
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 15 11:05:32 AM UTC 2026

System load: 0.11               Processes:                273
Usage of /:  49.5% of 9.75GB    Users logged in:         0
Memory usage: 21%              IPv4 address for ens33: 192.168.23.160
Swap usage:  0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

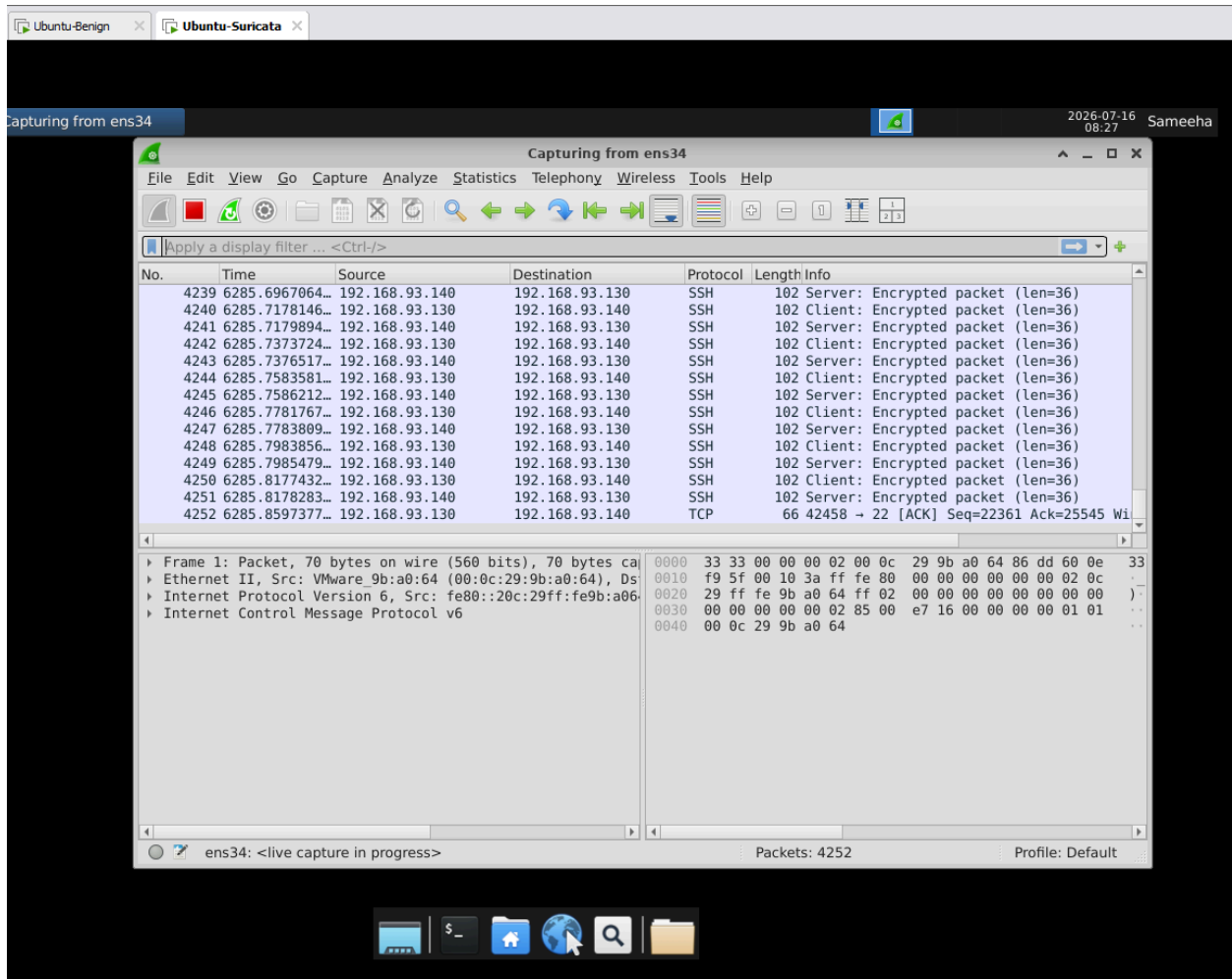
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

35 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ubuntu@ubuntu:~$
```



```
us@ubuntu:~$  
Session Actions Edit View Help  
ification: Attempted Administrator Privilege Gain [Priority: 1] {TCP} 192.168.93.130:47416 → 192.168.93.139:80  
07/16/2026-06:41:02.414002 [**] [1:2221015:1] SURICATA HTTP Host header ambiguous [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.414002 [**] [1:2221028:1] SURICATA HTTP Host header invalid [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.420903 [**] [1:2029206:5] ET EXPLOIT Possible Citrix Application Delivery Controller Arbitrary Code Execution Attempt (CVE-2019-19781) [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.420903 [**] [1:2035109:2] ET EXPLOIT Possible Citrix Application Delivery Controller Arbitrary Code Execution Attempt (CVE-2019-19781) M6 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.420903 [**] [1:2035110:2] ET EXPLOIT Citrix Application Delivery Controller Arbitrary Code Execution Attempt Scanner Attempt (CVE-2019-19781) [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.448158 [**] [1:2030469:7] ET EXPLOIT F5 TMUI RCE vulnerability CVE-2020-5902 Attempt M1 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.469034 [**] [1:2221010:1] SURICATA HTTP unable to match response to request [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.93.139:80 → 192.168.93.130:47418  
07/16/2026-06:41:02.503691 [**] [1:2030483:4] ET EXPLOIT F5 TMUI RCE vulnerability CVE-2020-5902 Attempt M2 [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.510332 [**] [1:2221015:1] SURICATA HTTP Host header ambiguous [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.93.130:47418 → 192.168.93.139:80  
07/16/2026-06:41:02.584095 [**] [1:2030585:1] ET EXPLOIT Cisco ASA/Firepower Unauthenticated File Read (CVE-2020-3452) M3 [**] [Classification: Attempted User Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47422 → 192.168.93.139:80  
07/16/2026-06:41:02.584095 [**] [1:2034262:1] ET EXPLOIT Cisco ASA and Firepower Path Traversal Vulnerability M1 (CVE-2020-3452) [**] [Classification: Attempted Administrator Privilege Gain] [Priority: 1] {TCP} 192.168.93.130:47422 → 192.168.93.139:80  
07/16/2026-06:42:36.921670 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.93.130:42626 → 192.168.93.139:80  
07/16/2026-06:42:36.923028 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.93.130:42632 → 192.168.93.139:80  
07/16/2026-06:42:36.928831 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.93.130:42634 → 192.168.93.139:80  
07/16/2026-06:42:36.930727 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.93.130:42648 → 192.168.93.139:80  
07/16/2026-06:47:20.023948 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-07:02:19.947190 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-07:17:17.649710 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-07:32:17.371275 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
07/16/2026-07:47:17.381690 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.93.130:68 → 192.168.93.254:67  
^C  
us@ubuntu:~$
```